

Solved selected problems of Modern Physics for Scientists and Engineers - John Taylor

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Chapter 7 - The Schrödinger Equation in One Dimension

Solution. 7.8 Equation (7.118) states that

$$y(x, t) = A \sin kx \cos \omega t$$

We know that $k = 2\pi/\lambda$ and $\omega = 2\pi/T$ hence the equation becomes

$$y(x, t) = A \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi t}{T}$$

Let us take a point $x = x_0 + \lambda$ with t fixed then

$$\begin{aligned} y(x_0 + \lambda, t) &= A \sin \left(\frac{2\pi(x_0 + \lambda)}{\lambda} \right) \cos \frac{2\pi t}{T} \\ &= A \sin \left(\frac{2\pi x_0}{\lambda} + 2\pi \right) \cos \frac{2\pi t}{T} \\ &= A \left[\sin \frac{2\pi x_0}{\lambda} \cos 2\pi + \cos \frac{2\pi x_0}{\lambda} \sin 2\pi \right] \cos \frac{2\pi t}{T} \\ &= A \left[\sin \frac{2\pi x_0}{\lambda} + 0 \right] \cos \frac{2\pi t}{T} \\ &= A \sin \frac{2\pi x_0}{\lambda} \cos \frac{2\pi t}{T} \\ &= y(x_0, t) \end{aligned}$$

Where we used the trigonometric identity $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$. This shows that the wave has the same value if we move from x_0 to a point $x_0 + \lambda$.

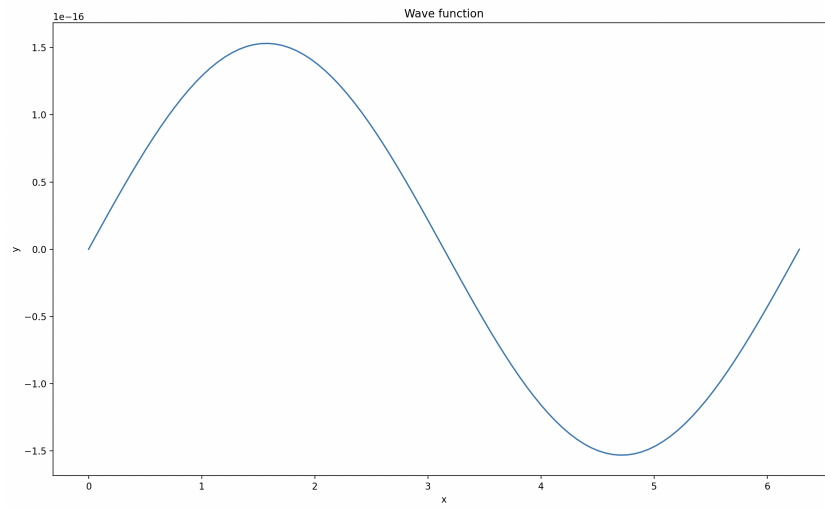
In the same way, let us take $t = t_0 + T$ with x fixed then

$$\begin{aligned}
 y(x, t_0 + T) &= A \sin \frac{2\pi x}{\lambda} \cos \left(\frac{2\pi(t_0 + T)}{T} \right) \\
 &= A \sin \frac{2\pi x}{\lambda} \cos \left(\frac{2\pi t_0}{T} + 2\pi \right) \\
 &= A \sin \frac{2\pi x}{\lambda} \left[\cos \frac{2\pi t_0}{T} \cos 2\pi - \sin \frac{2\pi t_0}{T} \sin 2\pi \right] \\
 &= A \sin \frac{2\pi x}{\lambda} \left[\cos \frac{2\pi t_0}{T} - 0 \right] \\
 &= A \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi t_0}{T} \\
 &= y(x, t_0)
 \end{aligned}$$

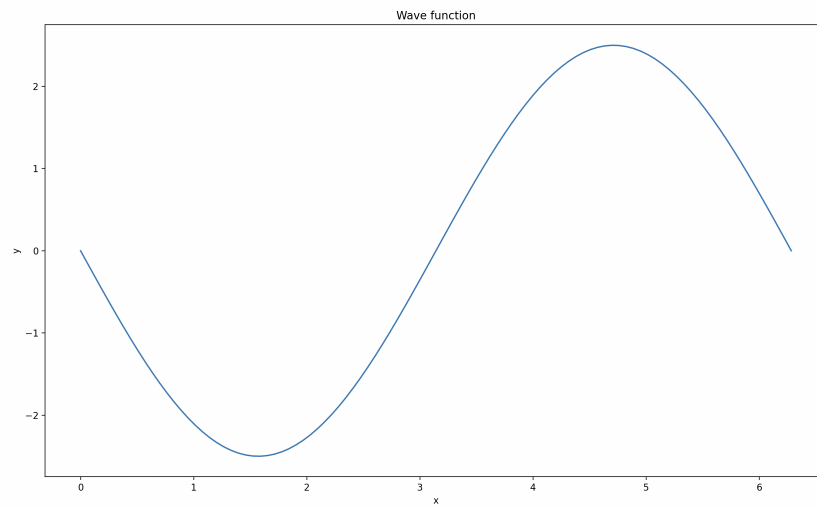
Where we used the trigonometric identity $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$. □

Solution. 7.9

(a) For $t = 0.05$ s we have that



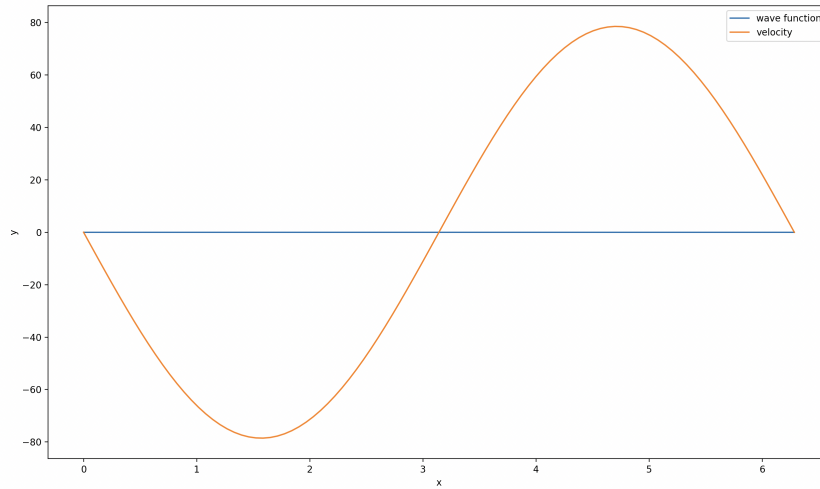
And for $t = 0.1$ s we get that



(b) Let x be fixed then the derivative of (7.118) with respect to t gives us

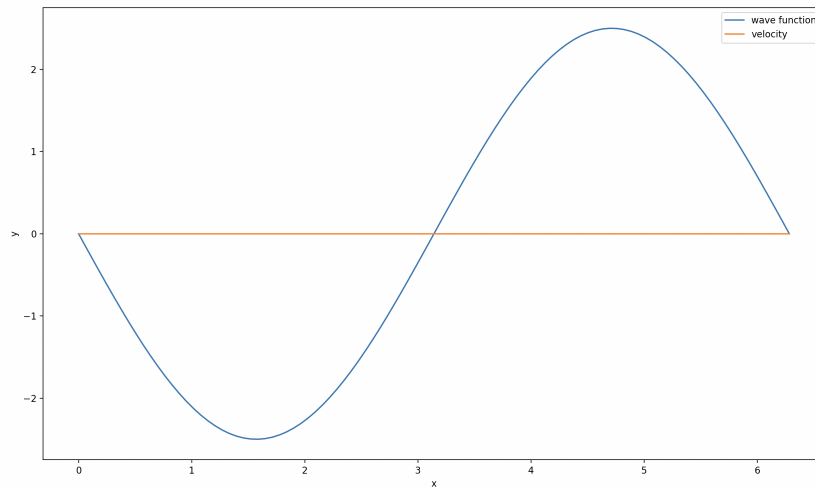
$$\frac{\partial y(x, t)}{\partial t} = -\omega A \sin kx \sin \omega t$$

Then for $t = 0.05 \text{ s}$ the transverse velocity is



So the velocity is high when the wave function amplitude is small.

Finally, for $t = 0.1 \text{ s}$ we get that

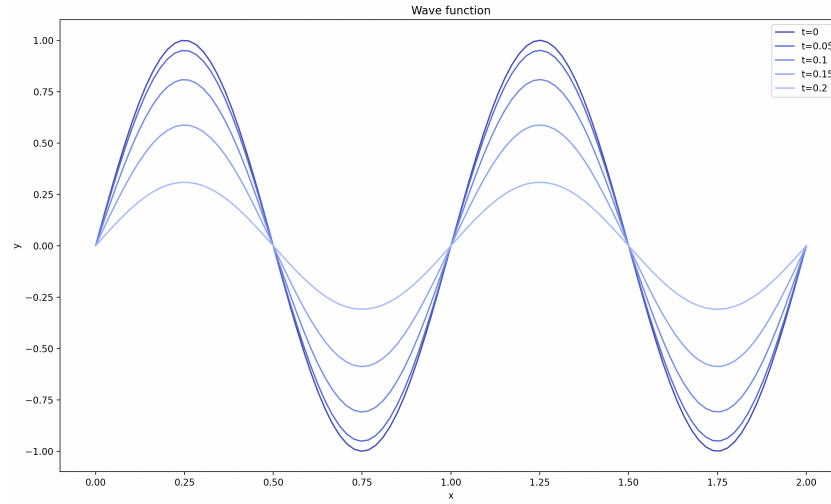


Here the velocity is small when the wave function amplitude is significant.

□

Solution. 7.60

The standing wave is at full amplitude ($A = 1$) at $t = 0$ and then decreases as times passes.

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Solution. 7.62 Executing `ch7-7.62.py` file the required animation can be generated.

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